

7.1 BASIC GOVERNING EQUATIONS FOR GROUNDWATER FLOW

Exam Questions Addressed:

- Explain the continuity equation used in groundwater flow analysis. [2071 Bhadra]

7.1 Continuity Equation in Groundwater Flow

The basic governing equation for groundwater flow is derived by combining the **Steady Continuity Equation** for incompressible flow with **Darcy's Law**.

1. Continuity Equation:

For steady, incompressible flow in three dimensions, the conservation of mass states that the net velocity divergence is zero:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

2. Darcy's Law:

Darcy's law states that the velocity through a porous media is directly proportional to the hydraulic gradient. For a homogeneous and isotropic formation:

$$u = -K \frac{\partial h}{\partial x}, \quad v = -K \frac{\partial h}{\partial y}, \quad w = -K \frac{\partial h}{\partial z}$$

Where h is the groundwater head (piezometric head) and K is the hydraulic conductivity.

3. Laplace Equation:

Substituting the Darcy velocity components back into the continuity equation yields the famous **Laplace Equation** for groundwater head. For 2D flow in the horizontal plane:

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = 0$$



7.2 NUMERICAL SCHEME FOR GROUNDWATER FLOW

Exam Questions Addressed:

- Derive finite difference scheme for 2D groundwater simulation in steady state for homogeneous & isotropic aquifer. [2076 Bhadra, 2075 Bhadra]
- Describe boundary conditions and flow coefficients. [2072 Ashwin]

7.2 Finite Difference Scheme for 2D Groundwater Flow

Consider an interior grid point (i, j) in a 2D unconfined aquifer. Let q represent the Darcy fluxes (per unit area) moving between adjacent grids, and q_F represent the net volumetric withdrawal (pumping) out of the grid.

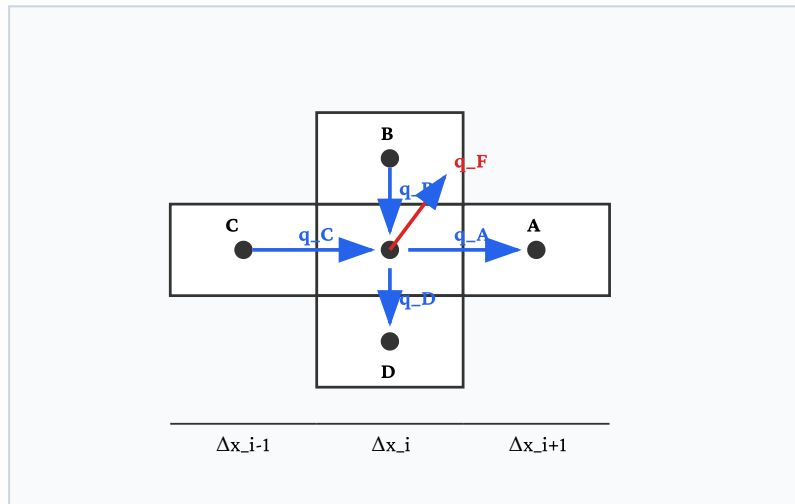


Figure: Grid definitions for 2D flow Darcy fluxes

1. Control Volume Continuity:

The principle of mass conservation states: *Inflow - Outflow = Rate of Change of Storage*.

$$\Delta x_i \Delta y_j [q_C - q_D - q_A + q_B - q_F] = S \frac{\partial \phi}{\partial t}$$

Where S = Storage coefficient, and ϕ = Potential head.

2. Flow Coefficients via Darcy's Equation:

The flow rate leaving grid i towards grid $i + 1$ (i.e., Q_A) is given by Darcy's law in terms of transmissivity (T):

$$Q_A = -T(\Delta y_j) \frac{\phi_{i+1,j} - \phi_{i,j}}{\Delta x_{i+1}/2 + \Delta x_i/2} = -2T(\Delta y_j) \frac{\phi_{i+1,j} - \phi_{i,j}}{\Delta x_{i+1} + \Delta x_i}$$

The flux per unit area is $q_A = -Q_A/(\Delta x_i \Delta y_j)$:

$$q_A = A_{i,j}(\phi_{i+1,j} - \phi_{i,j}) \quad \text{Where} \quad A_{i,j} = \frac{2T}{\Delta x_i(\Delta x_i + \Delta x_{i+1})}$$

Similarly, the other directional fluxes and their coefficients are:

$$q_C = C_{i,j}(\phi_{i-1,j} - \phi_{i,j}) \quad \text{Where} \quad C_{i,j} = \frac{2T}{\Delta x_i(\Delta x_i + \Delta x_{i-1})}$$

$$q_D = D_{i,j}(\phi_{i,j+1} - \phi_{i,j}) \quad \text{Where} \quad D_{i,j} = \frac{2T}{\Delta y_j(\Delta y_{j+1} + \Delta y_j)}$$

$$q_B = B_{i,j}(\phi_{i,j-1} - \phi_{i,j}) \quad \text{Where} \quad B_{i,j} = \frac{2T}{\Delta y_j(\Delta y_{j-1} + \Delta y_j)}$$

3. Final Discretized Equation:

Substituting the fluxes back into the continuity equation and collecting terms around $\phi_{i,j}$ gives the general implicit 2D finite difference equation:

$$A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1} - E\phi_{i,j} = S \frac{\partial \phi}{\partial t} + q_F$$

$$\text{Where } E = A + B + C + D.$$

Steady State, Homogeneous & Isotropic Aquifer:

For steady flow ($\partial \phi / \partial t = 0$), no pumping ($q_F = 0$), and a square isotropic grid ($\Delta x = \Delta y$, so $A = B = C = D$), the equation drastically simplifies to the average of the 4 neighbors:

$$\phi_{i+1,j} + \phi_{i,j-1} + \phi_{i-1,j} + \phi_{i,j+1} - 4\phi_{i,j} = 0$$

4. Boundary Conditions:

- **Impervious / Barrier Boundary:** No flow occurs across the boundary. The flow coefficient from that side is forced to zero (e.g., if the left side is a barrier, $C = 0$).
- **Constant Head (Equipotential) Boundary:** The head value ϕ is fixed at the boundary nodes (e.g., adjacent to a large river or lake).



7.3 SIMULATION OF SEEPAGE UNDER A DAM

Exam Questions Addressed:

- Draw a neat sketch presenting the FD grid showing boundary conditions. [2080 Chaitra]
- Develop a steady-state 2D model for the simulation of seepage under a dam. [2079 Shrawan, 2074 Bhadra]
- Describe the iterative procedure / write the algorithm for computing potential and seepage. [2075 Bhadra, 2071 Bhadra]

7.3 Steady State 2D Model for Dam Seepage

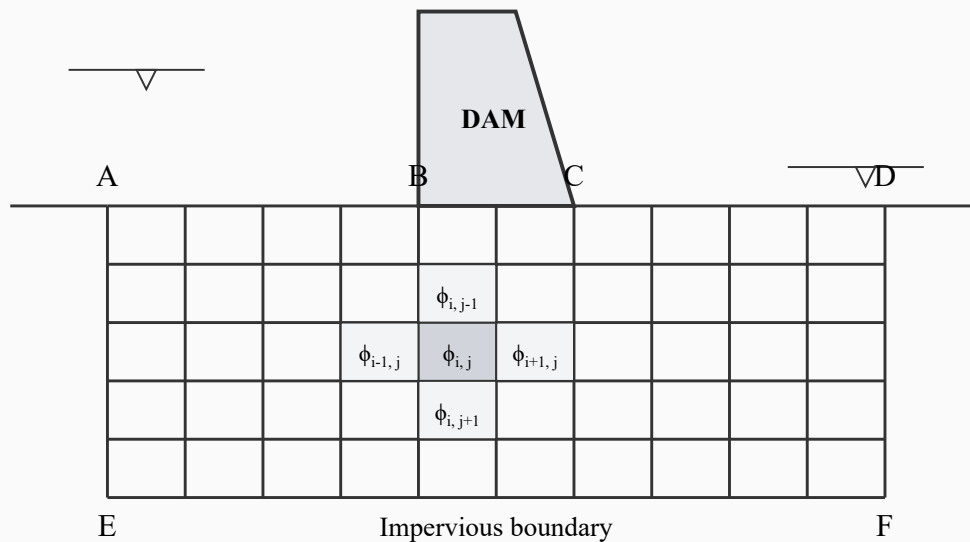


Figure: Finite difference grid for seepage under dam.

Based on the general 2D formulation derived in 7.2, the steady state potential head at any interior grid point (i, j) is given by:

$$\phi_{i,j}^+ = \frac{A\phi_{i+1,j} + B\phi_{i,j-1} + C\phi_{i-1,j} + D\phi_{i,j+1}}{E}$$

For a homogeneous and isotropic soil medium under the dam, this becomes:

$$\phi_{i,j}^+ = \frac{\phi_{i+1,j} + \phi_{i,j-1} + \phi_{i-1,j} + \phi_{i,j+1}}{4}$$

Boundary Constraints Identification:

- **AB and CD (Upstream/Downstream beds):** Constant Head boundaries (equipotential lines).
- **BC (Dam Base) and EF (Bottom Stratum):** Impervious boundaries (no vertical flow, $B = 0$ or $D = 0$ at these edges).
- **AE and DF (Left/Right limits):** Extended far enough to act as Barrier boundaries (no horizontal flow).

Iterative Algorithm for Computing Potential & Seepage:

1. **Initialize:** Establish the grid and apply the known constant head boundary conditions along AB and CD.
2. **Initial Guess:** Assign an initial estimated value (usually 0) to all interior unknown nodes $\phi_{i,j}$.
3. **Iterate:** Compute the new updated potential $\phi_{i,j}^+$ for every interior node using the surrounding 4 nodes (Gauss-Seidel method).
4. **Error Check:** Compute the absolute error $e = |\phi_{i,j}^+ - \phi_{i,j}|$ at every node.
5. **Convergence:** If the maximum error across the grid is greater than a specified tolerance, repeat steps 3 and 4. Otherwise, stop the iteration.
6. **Compute Seepage Rate:** Once the final potential field is converged, calculate the actual volumetric seepage using Darcy's Law across the grids:
Horizontal seepage: $Q_{i,j}^H = C_{i,j} \Delta x_i \Delta y_j (\phi_{i-1,j} - \phi_{i,j})$
Vertical seepage: $Q_{i,j}^V = B_{i,j} \Delta x_i \Delta y_j (\phi_{i,j-1} - \phi_{i,j})$



7.4 RIVER STAGE WATER TABLE INTERACTION

Exam Questions Addressed:

- Derive finite difference equation to evaluate river stage-water table interaction for 1D flow. [2079 Chaitra, 2070 Magh]
- Express in terms of / Develop a tridiagonal coefficient matrix. [2079 Jestha, 2073 Magh]
- Explain the 1D implicit model to evaluate the interaction. [2070 Bhadra]

7.4 1D Implicit Model for River-Aquifer Interaction

This model simulates the dynamic interaction of a river recharging a groundwater aquifer (or vice versa). The governing partial differential equation for 1D unsteady groundwater flow is:

$$\frac{\partial}{\partial x} \left(T \frac{\partial \phi}{\partial x} \right) = S \frac{\partial \phi}{\partial t} + q_F$$

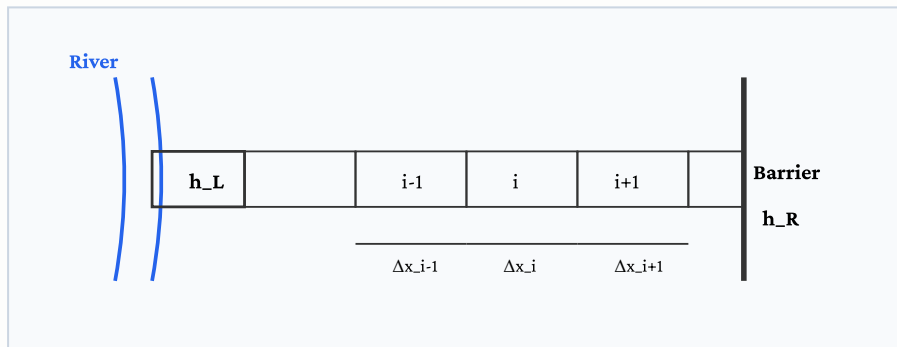


Figure: 1D Finite Difference Grid for River-Aquifer Interaction

By applying the continuity principle over a 1D grid space, the implicit finite difference formulation (evaluating spatial derivatives at the future time step $+$) for any internal grid i is:

$$A_i(h_{i+1}^+ - h_i^+) + C_i(h_{i-1}^+ - h_i^+) = \frac{S_i}{\Delta t}(h_i^+ - h_i) + q_F$$

Where h and h^+ represent the water table elevation at the beginning and end of time step Δt , respectively.

The spatial flow coefficients are defined as:

$$A_i = \frac{2T}{\Delta x_i(\Delta x_i + \Delta x_{i+1})} \quad \text{and} \quad C_i = \frac{2T}{\Delta x_i(\Delta x_i + \Delta x_{i-1})}$$

To simplify, we define lumped constants to collect the unknown future terms (h^+) on the LHS and known present terms (h) on the RHS:

$$E_i = - \left(A_i + C_i + \frac{S_i}{\Delta t} \right) \quad \text{and} \quad Q_i = - \frac{S_i}{\Delta t} h_i + q_F$$

This reduces the continuity equation to a simple linear algebraic equation:

$$C_i h_{i-1}^+ + E_i h_i^+ + A_i h_{i+1}^+ = Q_i$$

Tridiagonal Coefficient Matrix Formulation:

Writing this equation sequentially for every grid node from $i = 1$ to n results in a banded system of equations. Because each node only depends on its immediate left and right neighbors, it forms a **Tridiagonal Matrix**:

$$\begin{bmatrix} E_1 & A_1 & 0 & \dots & 0 \\ C_2 & E_2 & A_2 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & C_{n-1} & E_{n-1} & A_{n-1} \\ 0 & \dots & 0 & C_n & E_n \end{bmatrix} \begin{bmatrix} h_1^+ \\ h_2^+ \\ \vdots \\ h_{n-1}^+ \\ h_n^+ \end{bmatrix} = \begin{bmatrix} Q_1 - C_1 h_L^+ \\ Q_2 \\ \vdots \\ Q_{n-1} \\ Q_n - A_n h_R^+ \end{bmatrix}$$

Boundary Constraints: h_L^+ is the dynamic river stage (constant head boundary) at the left. If the right boundary is a physical barrier (no-flow), then the coefficient $A_n = 0$.

This matrix system is then efficiently solved at every time step using the Thomas Algorithm to find the new groundwater head profile. The interaction (loss or gain) at the river is simply the computed seepage entering or leaving grid element 1.